

Technology Stack

Date	15 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack:

Technology Stack Details

Architecture			
S.No	Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Used to design the user interface where customers enter their details and view credit card approval prediction results. Provides a simple and responsive user experience.
2	Backend / Server-Side	Python Flask 3.1.3	Flask is used to create the backend server and API endpoints. It connects the frontend with the machine learning model and handles prediction requests efficiently.
3	Database / Data Storage	Pandas / CSV Dataset / Joblib Model Storage	Pandas is used for data processing and handling datasets. Joblib is used to save and load the trained machine learning model for prediction.
4	Machine Learning & Data Processing	Scikit-learn 1.8.0, XGBoost 3.3.0, NumPy 2.2.4, Pandas 2.2.3	Scikit-learn provides ML algorithms and evaluation tools. XGBoost improves prediction accuracy using advanced boosting techniques. NumPy and Pandas support numerical computation and data preprocessing.
5	Cloud / Hosting / Deployment	Gunicorn 23.0.0	Gunicorn acts as a production WSGI server to deploy and run the Flask application reliably on cloud platforms.
6	Version Control & CI/CD	GitHub	GitHub is used for source code management, collaboration, version tracking, and maintaining project history.
7	Data Visualization / Analysis Tools	Matplotlib 3.10.8, Seaborn 0.13.2	Used for exploratory data analysis, visualization of patterns, and understanding factors affecting credit card approval predictions.

